CS4800: Designing Sustainable ICT Systems

Lecture 5: Sustainable Cloud Computing

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What will you learn in this lecture?

Week 1.6: Sustainable Cloud Computing

- Data center architecture: computing, cooling, power
- Environmental footprint of cloud computing, measurement methods
- Industry trends: compute growth, efficiency gains, energy
- Prospective solutions: cloud strategy, demand response, cooling technologies, waste heat reuse

What is a data center?

Definitions: data centers

- Data center: space that contains ICT equipment (from closet-sized server to >10,000 sqm)
- Classified by power load: used to be ~100 kW, now >100 MW, soon >1 GW
- Utilization rate: percentage of time that computers are in operation (<50% for enterprise loads, >80% for AI training)

Data center: a lot of computers in a big warehouse

Definitions: industry metrics

The Green Grid: industry consortium focusing on resource efficiency

- Power usage efficiency (PUE): ratio of total energy consumption, energy used for computing

$$\text{PUE} = \frac{\text{Total facility energy } [kWh]}{\text{IT equipment energy } [kWh]} > 1$$

- Water usage efficiency (WUE): ratio of water consumption, energy used for computing

$$\text{WUE} = \frac{\text{Annual water usage}}{\text{IT equipment energy}} \quad [L/kWh]$$

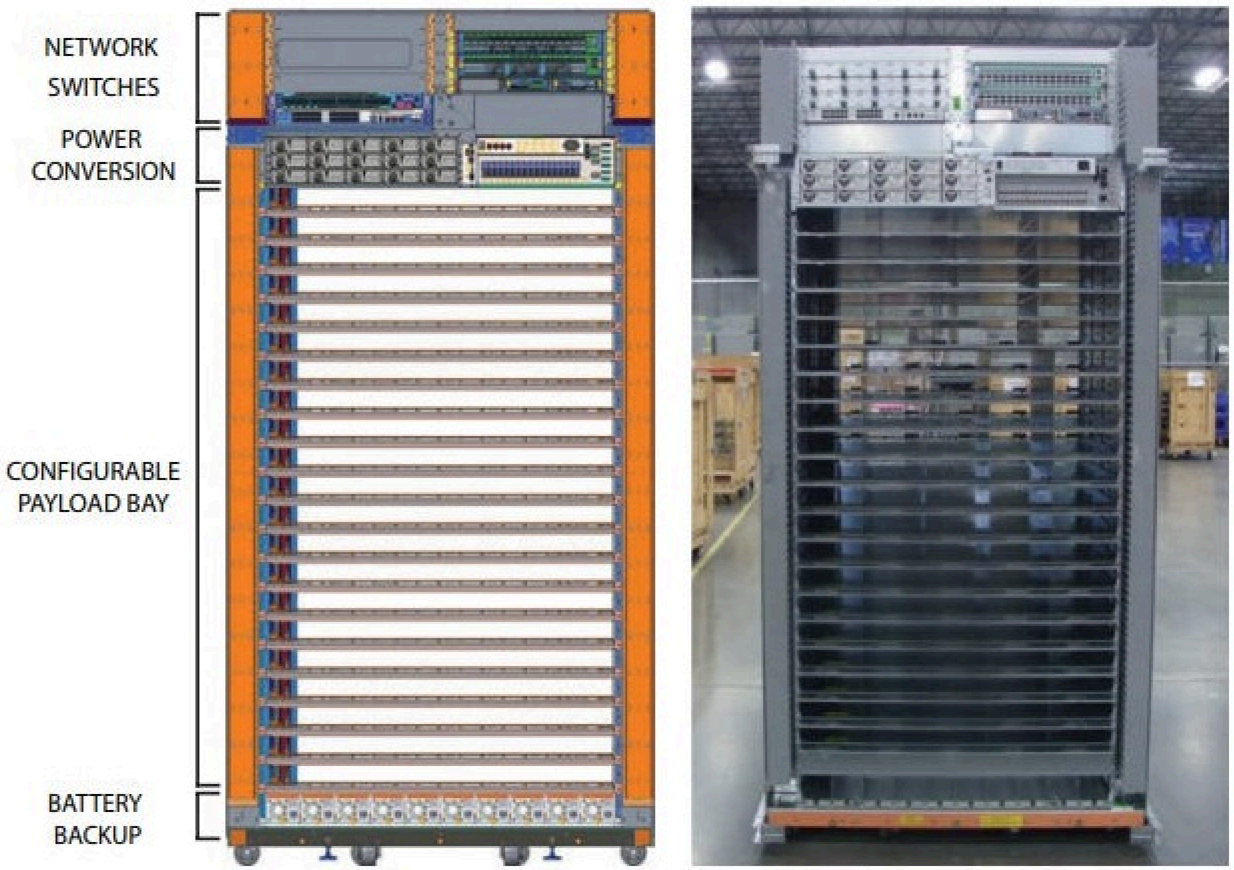
Uptime Institute: industry consortium focusing on data center reliability

- Reliability: four tiers (I-IV); higher tiers more reliable; Tier IV: theoretical uptime of 99.995%

How does a data center work?

Data center architecture: computing

- Up to tens of thousands of servers in hyperscale data centers
- Computers (servers) stored vertically in racks; several racks in a data hall
- Each server contains CPUs, GPUs, power electronics
- Also data storage and internal/ external data transfer networks
- Trade-off: increase server density to use copper rather than fiber optic cables, but increase need for cooling

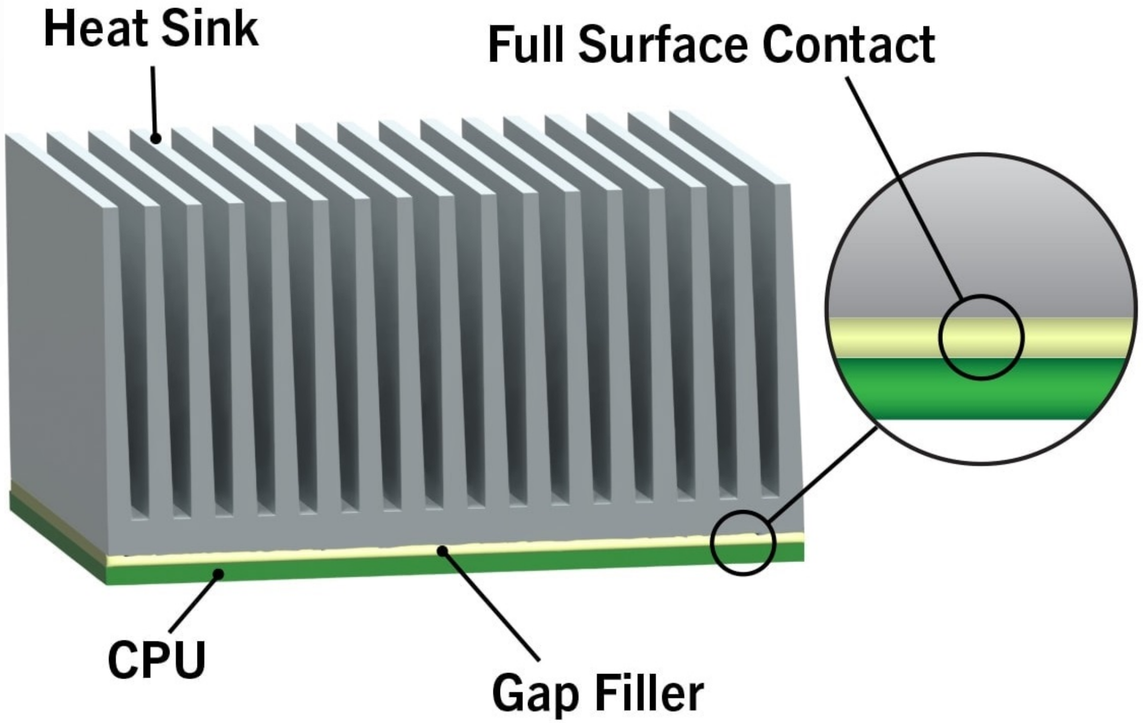


Server rack architecture

Data center architecture: computing

- Processor: uses electrical power to perform computations, producing heat

- Chips designed for thermal design power (TDP): maximum allowable heat
- Trend: increase in TDP, modern chips reaching 1500W
- Chip connected to thermal interface material (TIM), transfers heat to heat sink
- Higher TDP → larger heat sink, more cooling required

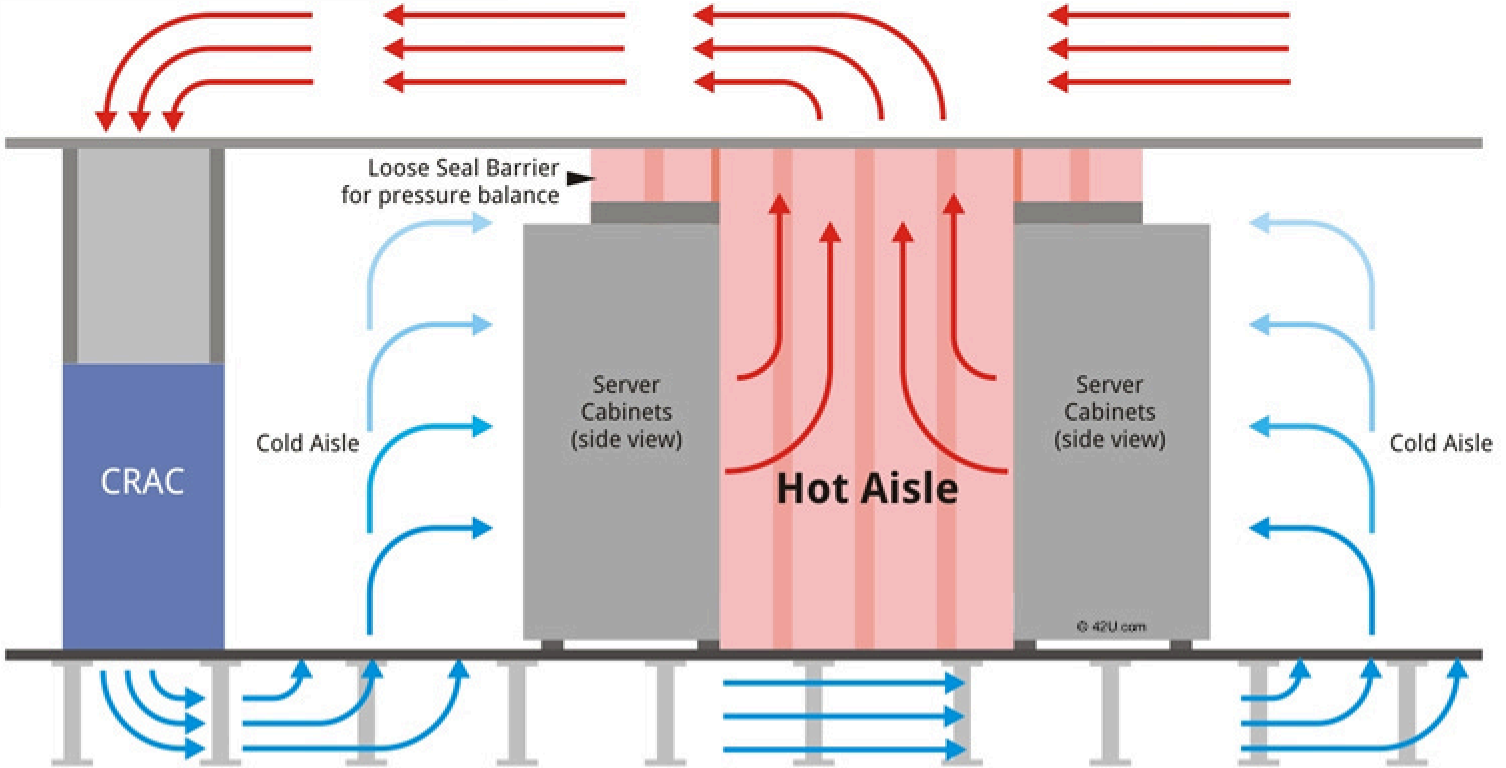


Thermal interface material and heat sink

Data center architecture: cooling

- Server fans evacuate heat to data hall
- [Computer room air conditioner/handler]() (CRAC/CRAH) in each data hall blows hot air over cold water coil, transferring heat; cold air recirculated to data hall

- Modern data centers: fan walls, rear-door heat exchangers (RDHx) for each rack
- Avoid [mixing]() between recirculated cold air and exhausted hot air → airflow management: aisle containment, computational fluid dynamics (CFD)

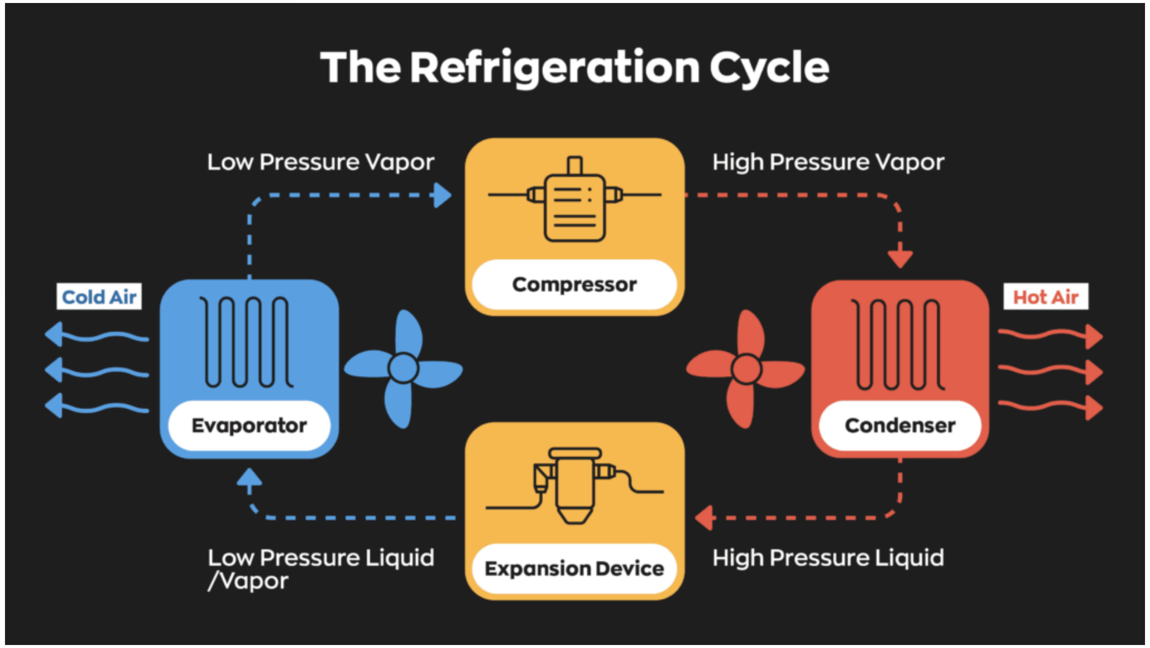


[Example: hot aisle containment]()

Data center architecture: cooling

- Warm water from CRAC/CRAH transferred to chiller, cooled down, and recirculated

- Chiller: large, energy-intensive, performs refrigeration cycle:
  1. Liquid refrigerant absorbs heat from water, evaporates
  2. Refrigerant vapor increases in pressure, temperature in compressor
  3. Refrigerant vapor transfers heat to outdoor cooling loop in condenser
  4. Expansion valve reduces pressure, temperature



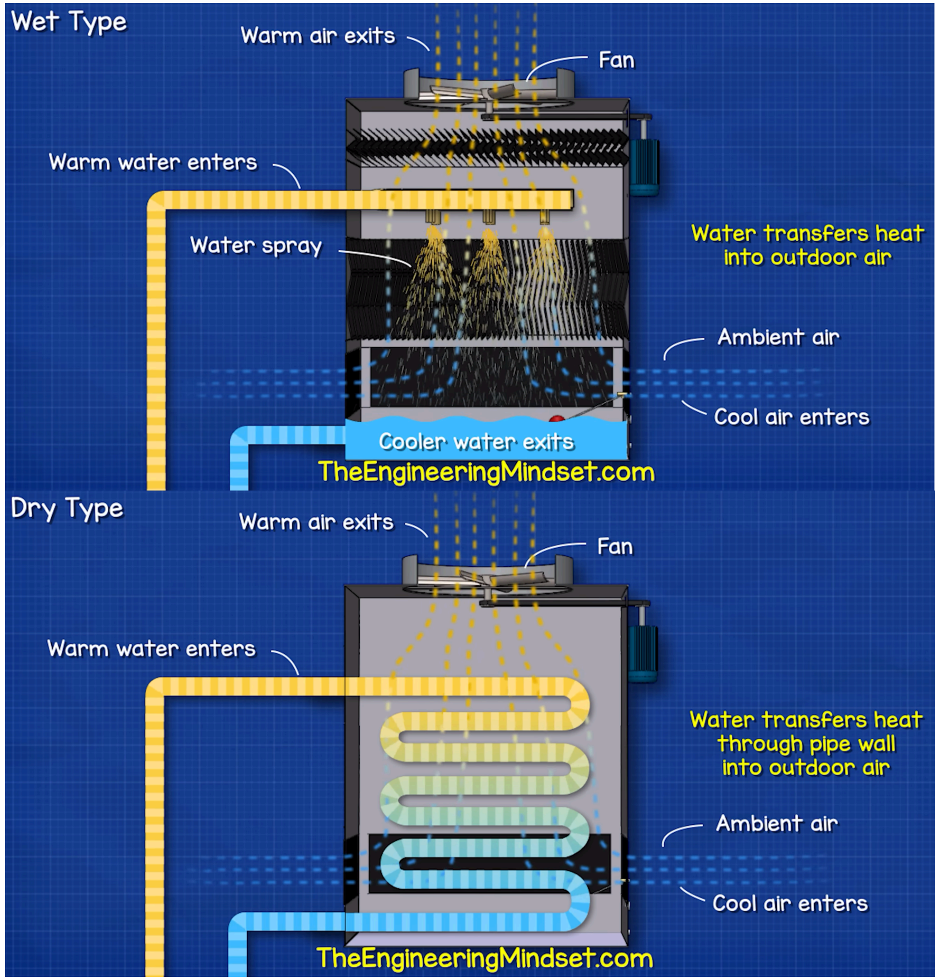
Four-step refrigeration cycle

Data center architecture: cooling

Heat moved from chiller to external cooling towers:

1. Wet cooling: water spray, evaporative cooling
   - High cooling capacity
   - Needs input water
   - Most effective: dry regions, low background humidity
2. Dry cooling: closed-loop, water flows through coil, cooled by fan
   - Lower cooling capacity
   - Consumes more electricity
   - Most effective: humid regions

Trade-off between energy and water efficiency!



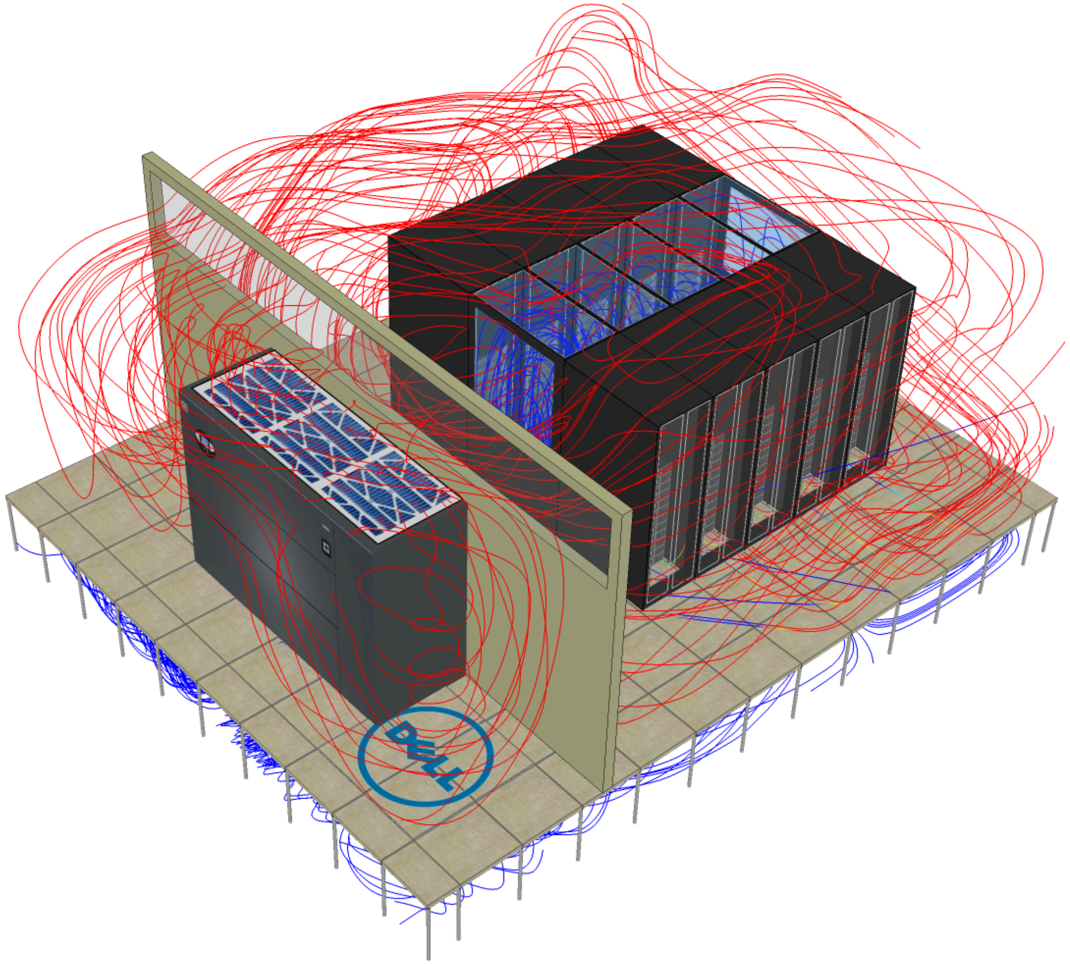
Wet and dry cooling towers

Data center architecture: cooling

Cooling energy efficiency gains: low hanging fruit already picked

- Servers designed for up to 45 degree Celsius operation
- Larger fans: same airflow, less input power
- Optimized, contained airflow
- Free cooling: outside environment used to reduce need for cooling
- Multiple cooling loops, heat storage tanks

Takeaway: cooling system is complex and energy-intensive, but absolutely necessary; without it, twenty minutes until total heat death



CFD modeling of data center airflows

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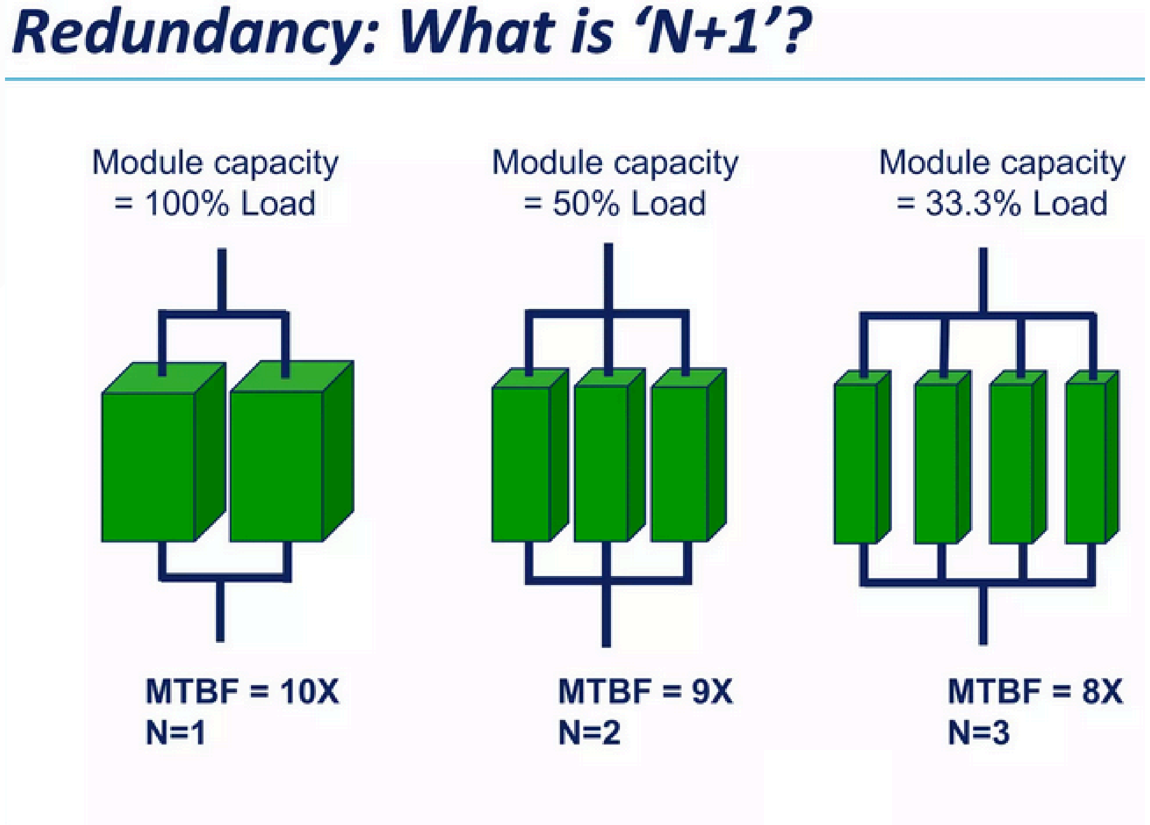
Data center architecture: power

- Sophisticated, redundant infrastructure delivers constant stable electricity to each server
- Power loss as function of current, voltage:

$$\mathrm{P} = \mathrm{I}\ [A] \cdot \mathrm{U}\ [V] = \mathrm{I}^2\ [A^2] \cdot \mathrm{R}\ [\Omega] \quad [W]$$

- Losses scale with current squared → keep current low, voltage high until data hall
- High voltage (HV) electricity from grid converted to medium voltage (MV) by power transformers
- Transformers in redundant N+1 configuration, partial operation (unused transformers deteriorate)



Example: redundant transformer architecture

Data center architecture: power

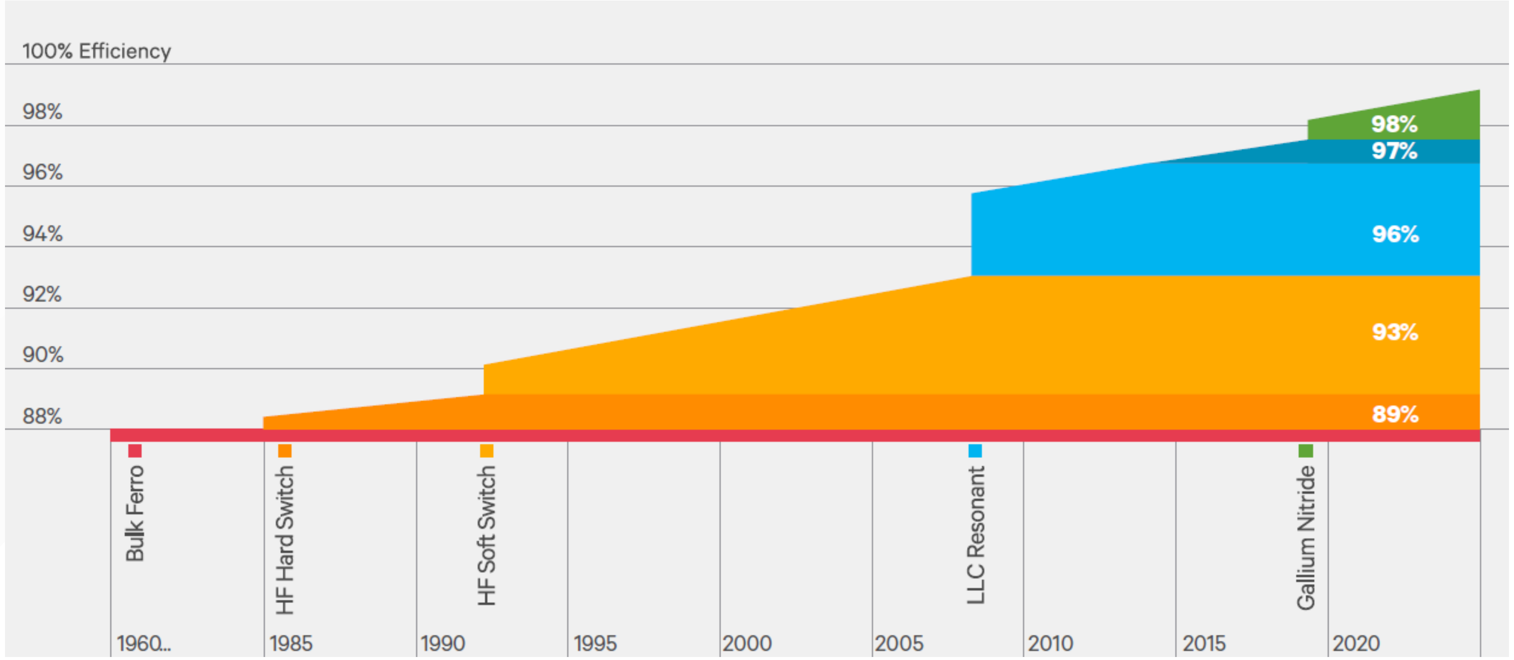
- From transformers, uninterruptible power supply (UPS) to servers, and power to cooling system
- UPS distributes power to each data hall, data pods with dedicated switchgear, generator
- Switchgear distributes, protects, meters power
- Diesel generator with 24-48 hours of fuel for outages
- Downstream from switchgear: battery bank, takes over instantly during outages while generators start up
- Rack level: overhead busway/flexible power cables, two redundant power distribution units (PDU) to servers

Backup diesel generators in a data center

Data center architecture: power

Power transmission efficiency gains: low hanging fruit already picked

- Power equipment more efficient: AC-DC rectifiers close to theoretical limit
- Idle chips ramp down
- Smaller transistors have higher compute efficiency
- Custom rack designs reduce required battery capacity



Efficiency trends for modern rectifiers

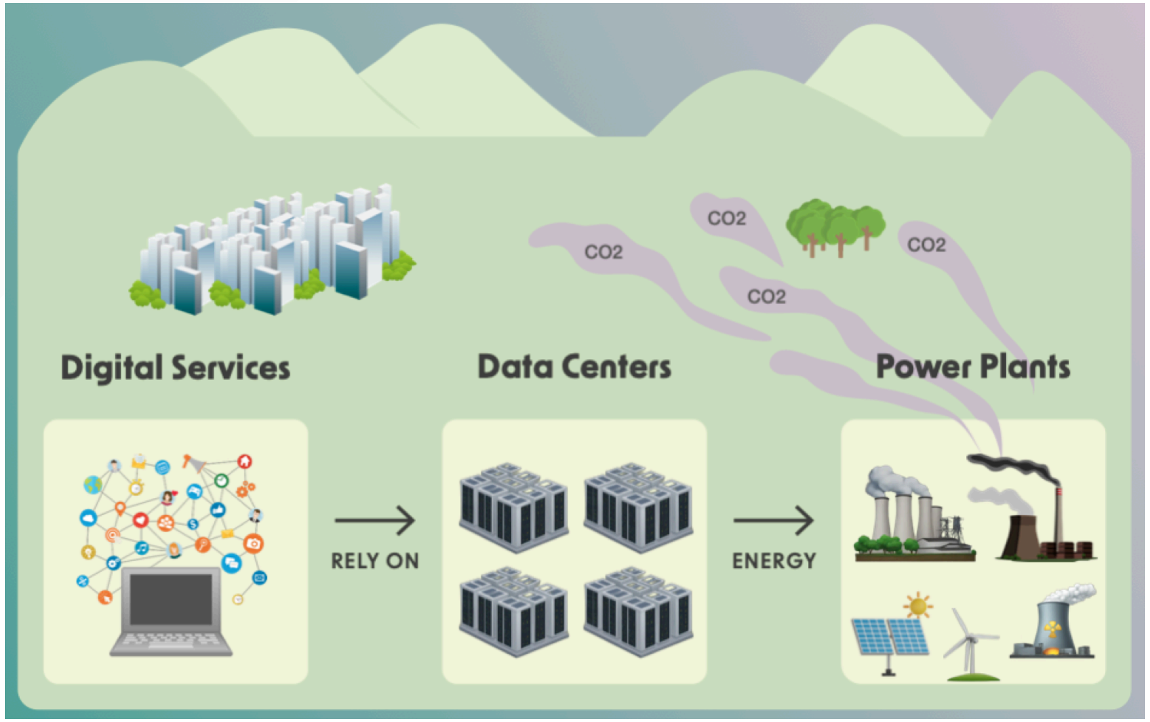
What are the impacts of the data center industry?

Data center impacts

- Data center industry has a significant energy/carbon/water/material footprint, affects local communities
- In this lecture: not just numbers, but also how to measure impacts
- Industry is remarkably opaque with e.g. power consumption → researchers rely on estimates
- Need to evaluate assumptions and methodologies critically!

Data center impacts: carbon footprint

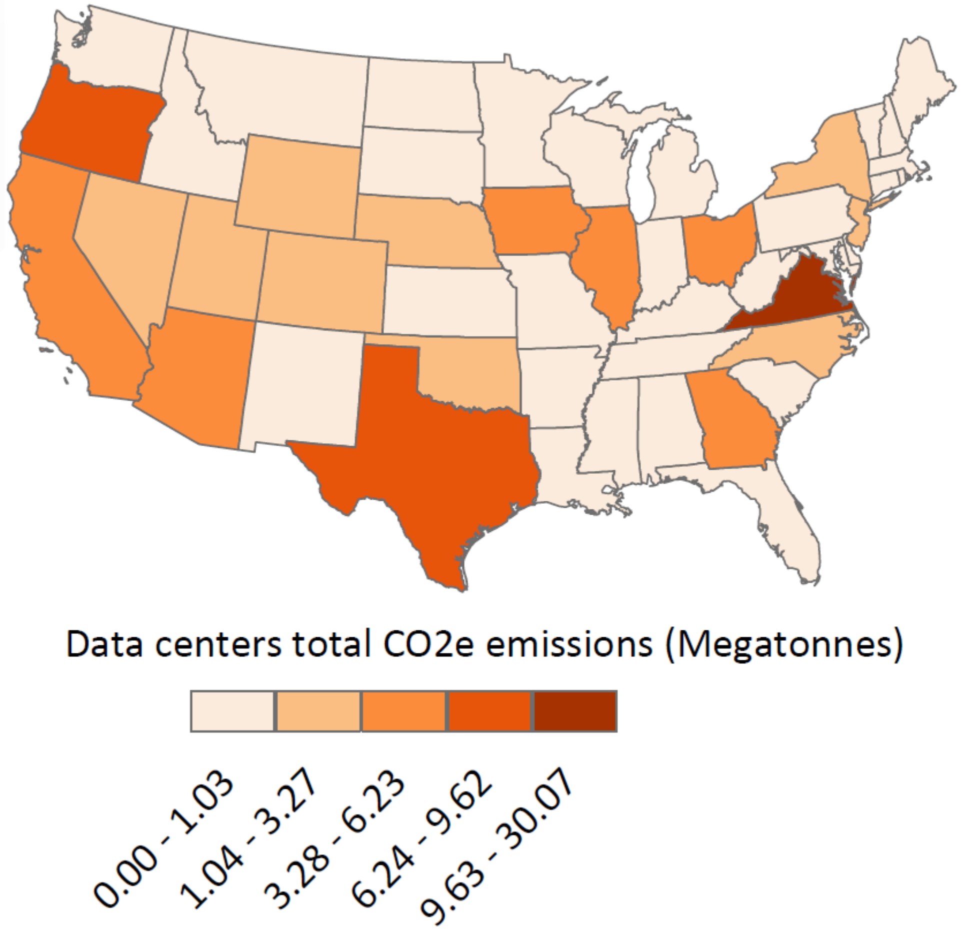
- Alex de Vries: 3Wh (10.8kJ) per AI search-related inference
- Shift Project: 3.7% of global GHG emissions caused by digital technologies
- Factors affecting data center carbon footprint:
  - Operational: electricity consumption
  - Operational: water consumption
  - Embedded: equipment lifetime



Data center emissions result from their electricity consumption

Data center impacts: carbon footprint

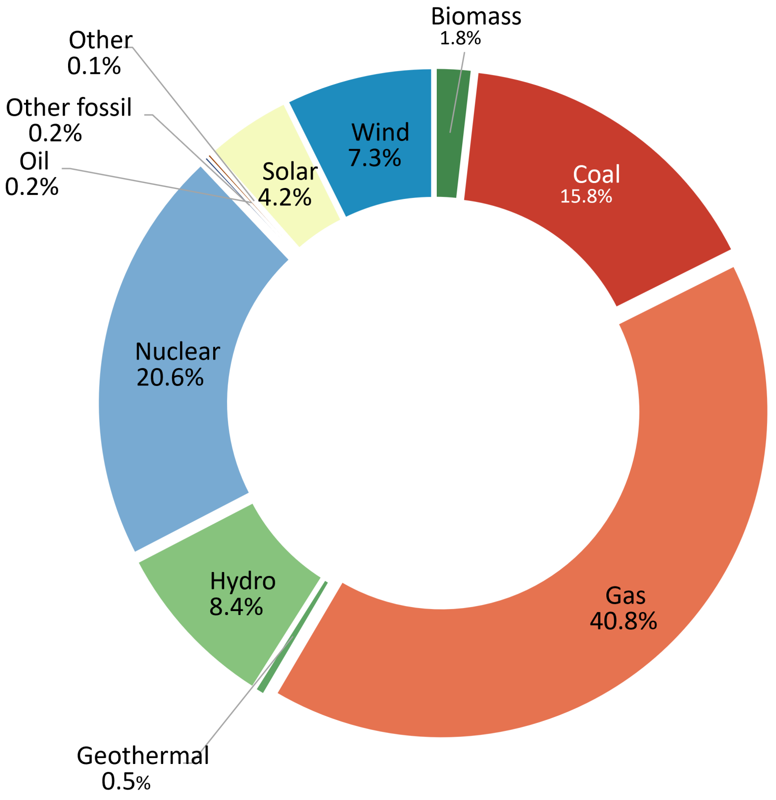
- Example: large-scale assessment (2024) of environmental impact of US data centers
- Methodology:
  1. Identified individual data center demand
  2. Found which power plants supplied each data center
  3. Estimated fuel mix for each power plant
  4. Attributed power plant emissions to each data center
  5. Made a neat map
- Result: data centers consumed >5% of 2022 US energy production



US data center emissions by state

Data center impacts: carbon footprint

- Average carbon intensity: 548 gCO_2-eq/kWh, 50% higher than US average → data centers in fossil-heavy areas
- Fossil fuels account for >50% of energy consumed
- Attributional methodology: external data used, results attributed to each data center → assumptions made:
  - Assumed 0.75 utilization
  - No differentiation between colocation/ hyperscale
  - Power capacity estimated by size where data not available
  - Only operational carbon footprint calculated, bill of materials not available
  - Are these assumptions reasonable?

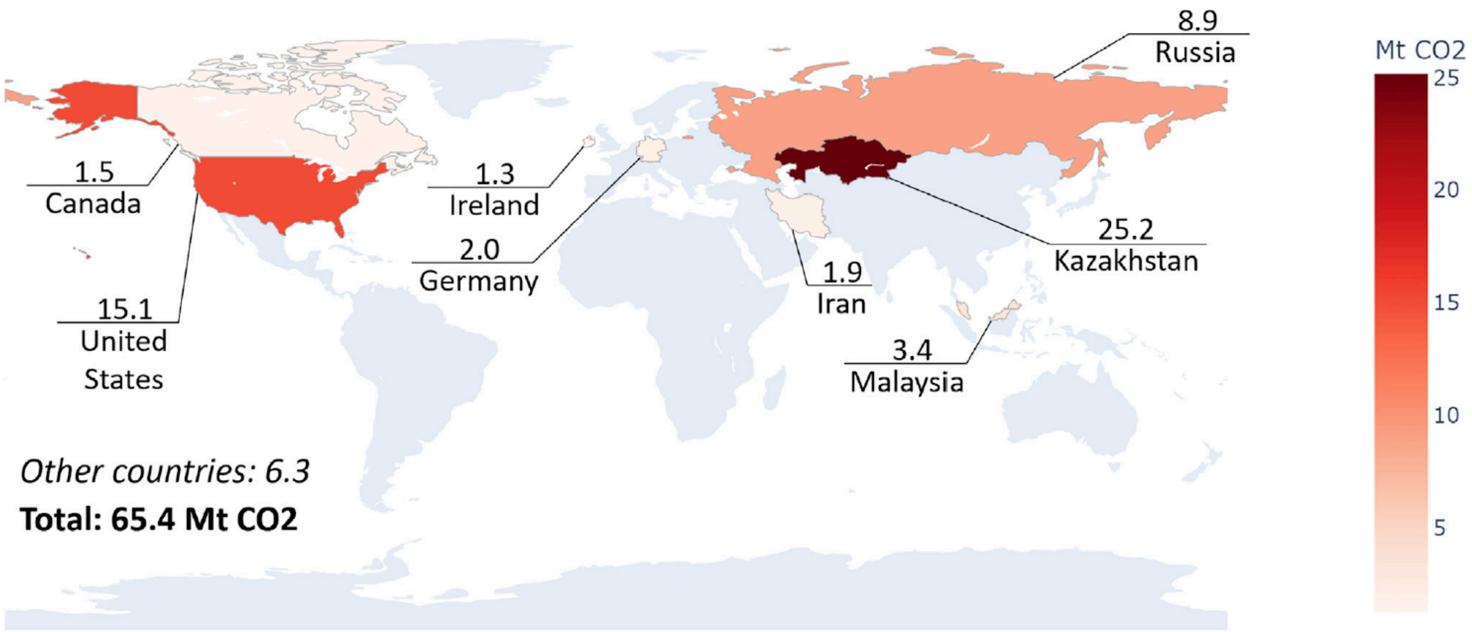


Estimated energy mix of US data centers

Data center impacts: carbon footprint

- Similar methodology (2025) applied to estimate carbon footprint of Bitcoin mining
- Miners reveal IP address → establish location, calculate carbon intensity of local electricity

- Global study (2021): carbon footprint of Bitcoin mining comparable to national emissions of Greece
- 2019: mining banned in China → miners migrated to Kazakhstan, US → more gas plants used



Global Bitcoin mining carbon footprint in megatons

Data center impacts: carbon footprint

- Cloud Carbon Footprint (CCF): open-source project for companies to calculate their cloud services carbon footprint
- Converts cloud provider usage data (compute, storage, networking) into energy consumption, estimates provider's data center PUE and carbon intensity:

$$1.\ \mathrm{CO_2eq_{total}} = \text{operational emissions} + \text{embedded emissions}$$

$$2.\ \text{operational emissions} = \text{cloud service usage}\ [use] \cdot \text{cloud energy conversion factors}\ [kWh/use] \cdot \text{cloud provider PUE}\ [-] \cdot \text{grid emissions factors}\ [metric\ tons\ CO_2eq]$$

$$3.\ \text{embedded emissions} = \text{estimated } \mathrm{CO_2eq} \text{ emissions from server manufacturing}$$

- Various estimates for grid emissions factors, manufacturing-related emissions used → inexact results, but still useful

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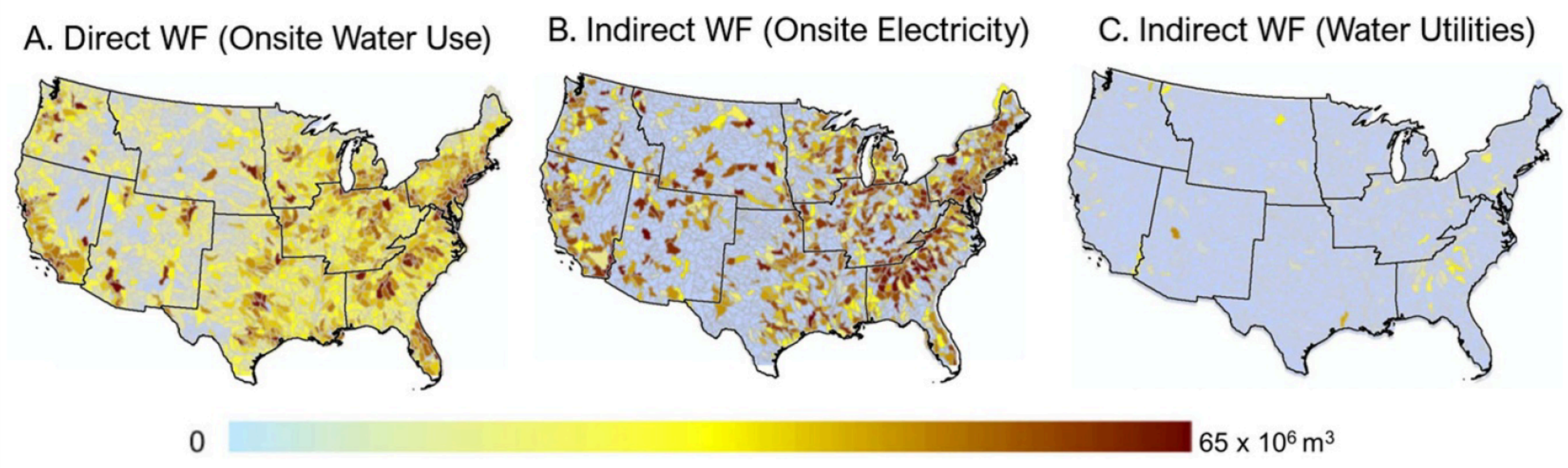
Data center impacts: water footprint

- US estimates: data centers among top ten water-consuming activities, 20% draw water from already stressed sources
- Latin America: significant recent buildout, including drought-vulnerable Mexico City
- Cloud providers do not report water consumption/local water stress levels → rely on estimates



Direct and indirect 2018 water footprint of US data centers in cubic meters

Data center impacts: water footprint

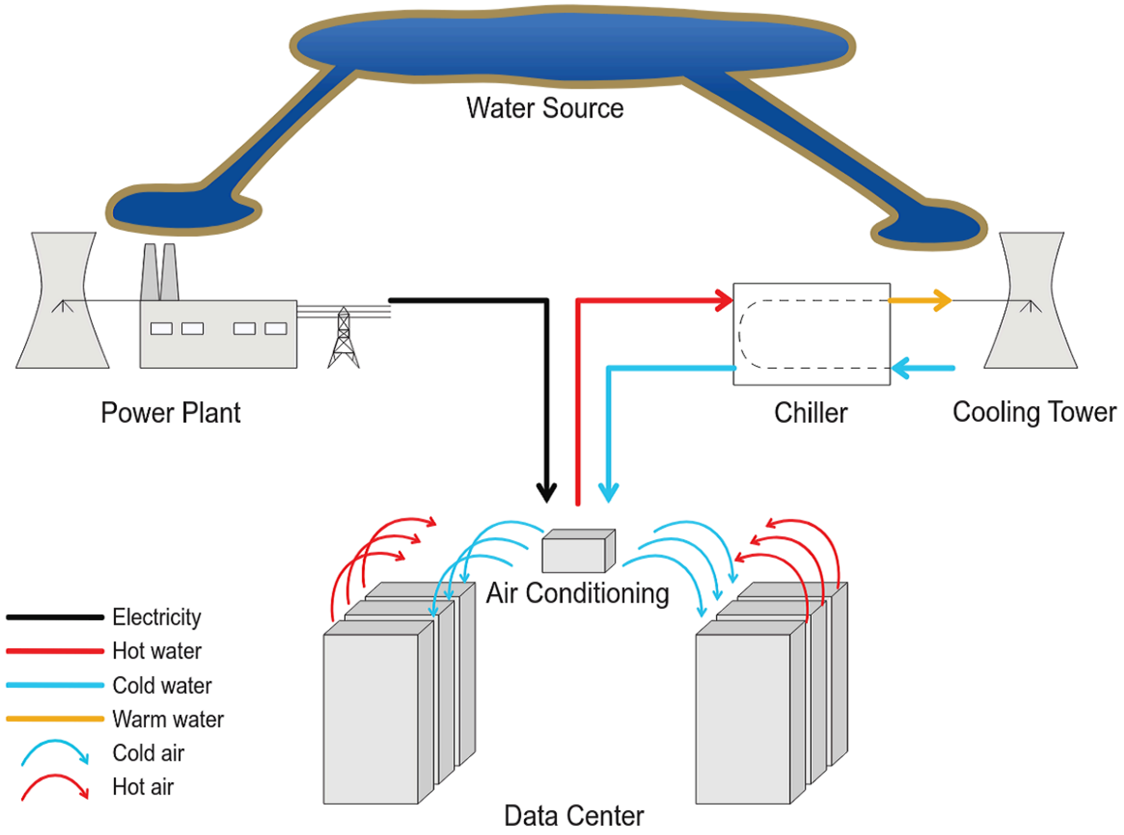
- Methodology to estimate water consumption of compute job M during time period t: sum of on-site water (for cooling) and off-site water (for energy generation):

$$W(M) = W_{\text{on}}(M) + W_{\text{off}}(M)$$

- On-site water consumption: function of energy used, data center WUE during job:

$$W_{\text{on}}(M) = \sum_{t=1}^{T} e(M,t) \cdot WUE_{\text{on}}(t)$$

- WUE estimated from external environment temperature: higher temperatures → more water needed for cooling



Data center water flows

Data center impacts: water footprint

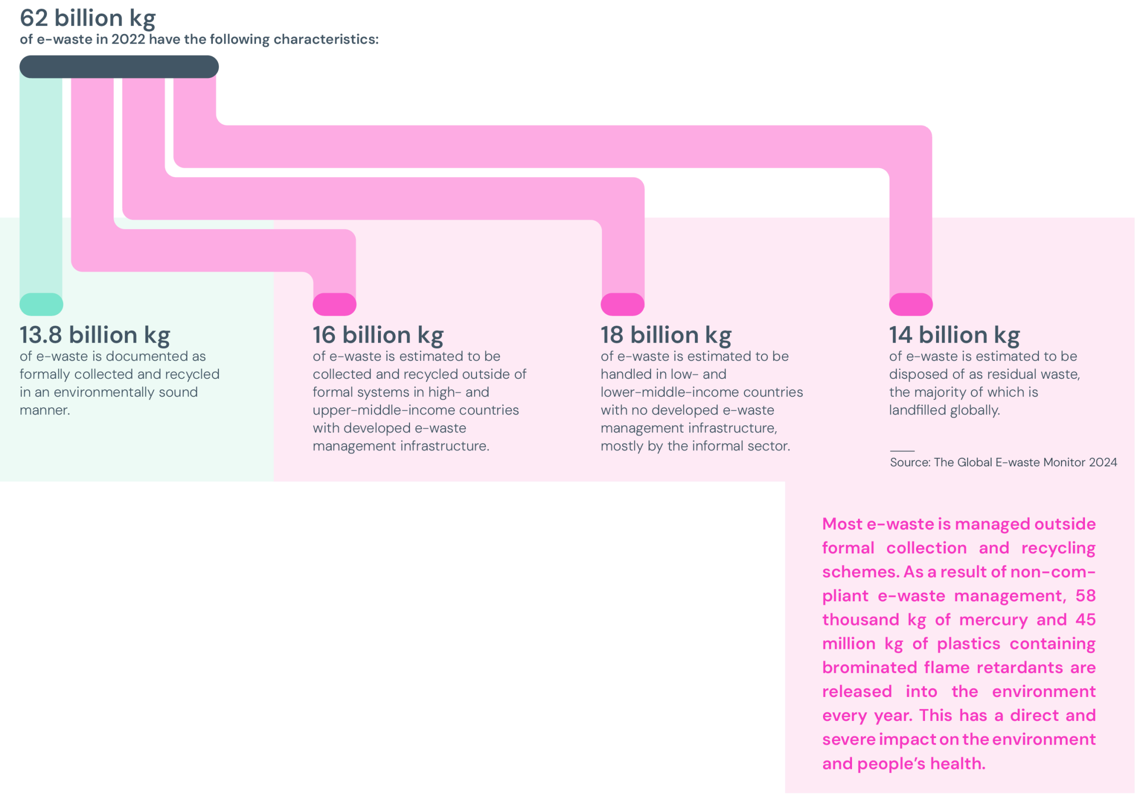
- Off-site water consumption: calculated as function of energy used, WUE (of power plant), and PUE (energy is also provided for cooling):

$$W_{\text{off}}(M) = \sum_{t=1}^{T} e(M,t) \cdot PUE(t) \cdot WUE_{\text{off}}(t)$$

- Power plant WUE: estimated based on fuel mix, relative water intensity factor (how much water required for 1kWh of energy)
- Takeaway: inherent operational trade-offs (e.g. using solar energy reduces CO_2 emissions, but works only during daytime, when temperatures are high, more cooling is needed)
- Some perspective: how does data center water use compare to agriculture? (hint: it doesn't)

Data center impacts: material footprint

- Entire frameworks developed to calculate ICT equipment embedded emissions
- Lecture 6: broad range of substances in electronics → toxic byproducts from processing
- No public data about server lifetime (estimated 3-5 years),
- Computing equipment usually not recycled → ends up in potentially toxic global waste stream (2022: 62 megatons)
- Also electrical, cooling systems, structural → up to 30% of lifetime emissions embedded in physical infrastructure

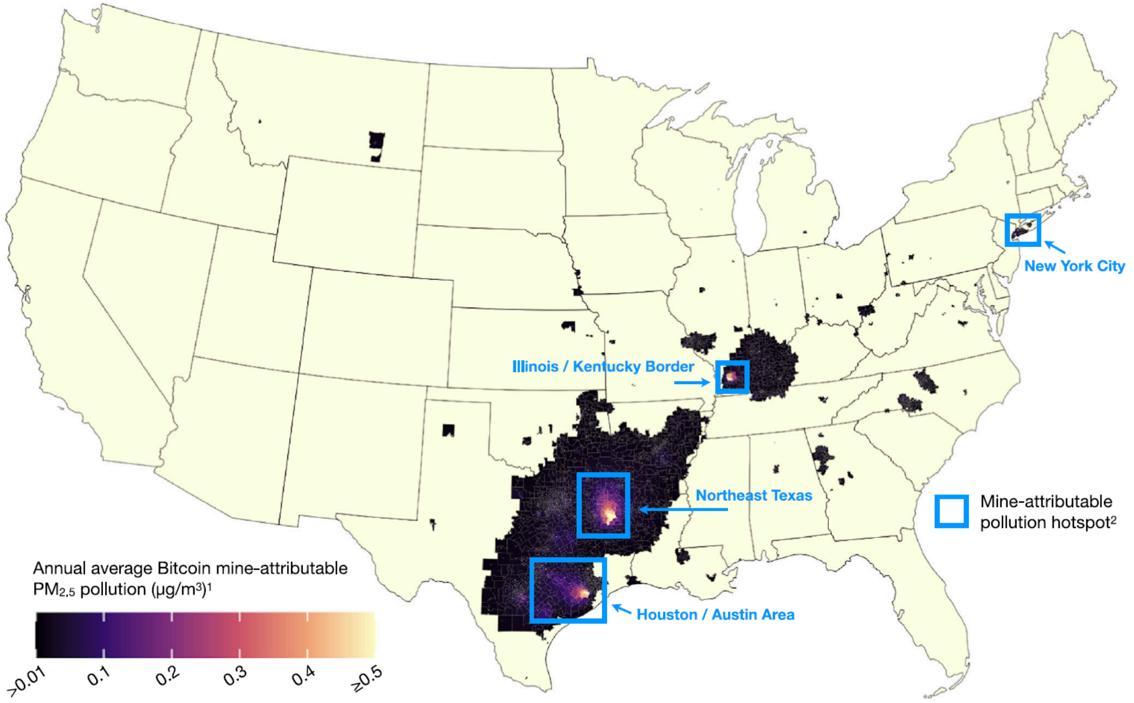


Global e-waste stream and recycled fraction

Data center impacts: local communities

- Most energy powering data centers comes from fossil fuels
- Coal, gas plants produce fine air pollution e.g. health-threatening $PM_{2.5}$

- Bitcoin mining-attributed $PM_{2.5}$ pollution: 46 million American citizens in 27 states were found to be exposed to measurable concentrations in 2022-2023
- Unchecked water consumption for data center cooling can deplete local aquifers → decreased access for local agriculture, amenities, household activities



Areas affected by $PM_{2.5}$ pollution caused by bitcoin mining

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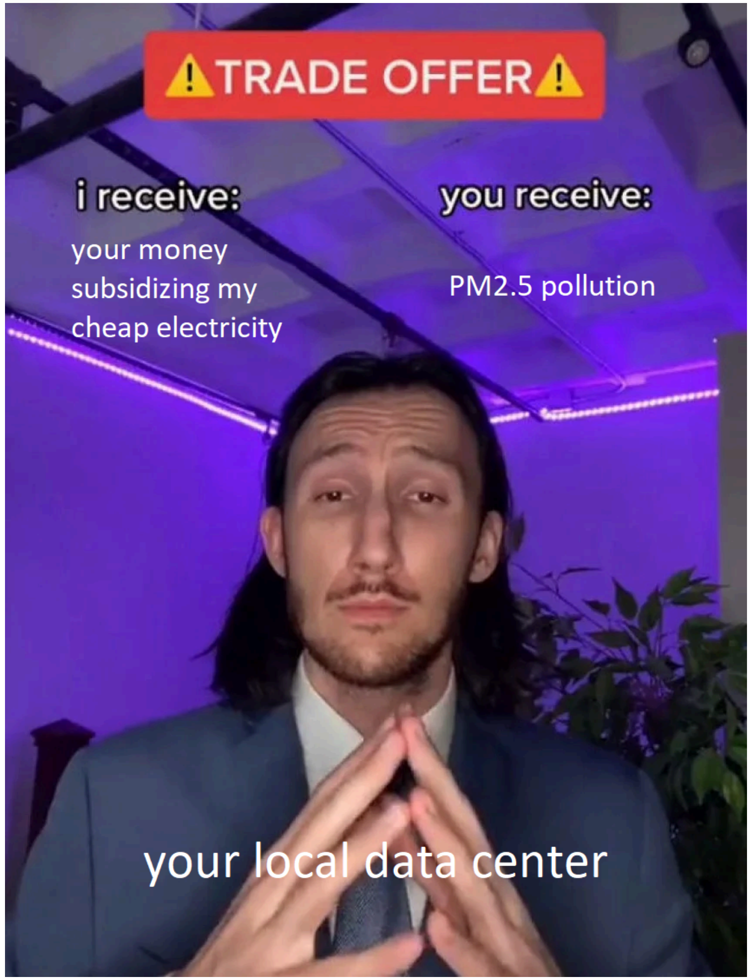
Data center impacts: local communities

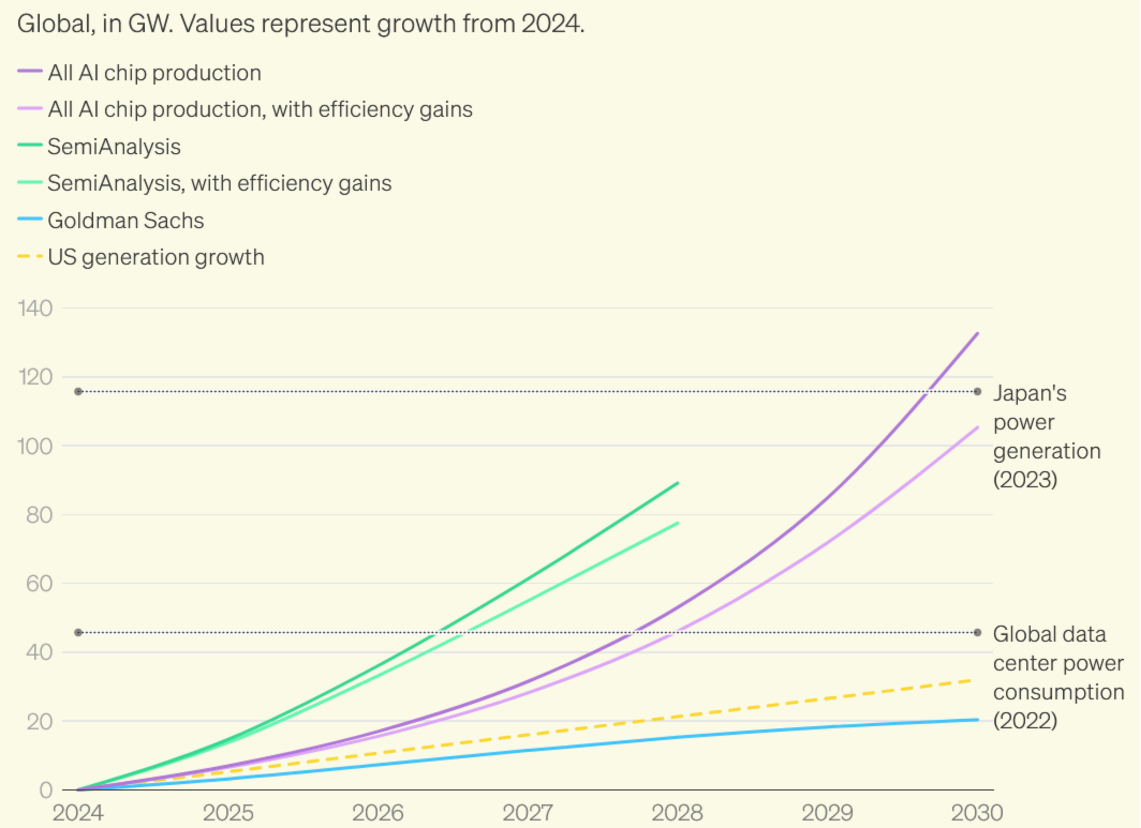
- Utility companies include costs of grid expansion in residential/industrial electricity prices
- Data centers offer stable, long-term income → utilities offer sweet deals to data centers, shift infrastructure costs to regular consumers
- Data centers eat up power generation from local low-cost energy sources with preferential agreements → cost of electricity increases for average consumer (you)



What are the important trends of the data center industry?

Industry trends: power demand

- All estimates: projected increase in global data center power demand: both AI and non-AI (doubled 2019-2024!)
- Goldman Sachs:
    - non-crypto data center power demand grows 80-240% by 2030
    - 3% of US power demand → 8%
    - 20% of growth is AI
- Other estimates (e.g. from expected production of AI chips): sharper, exponential growth

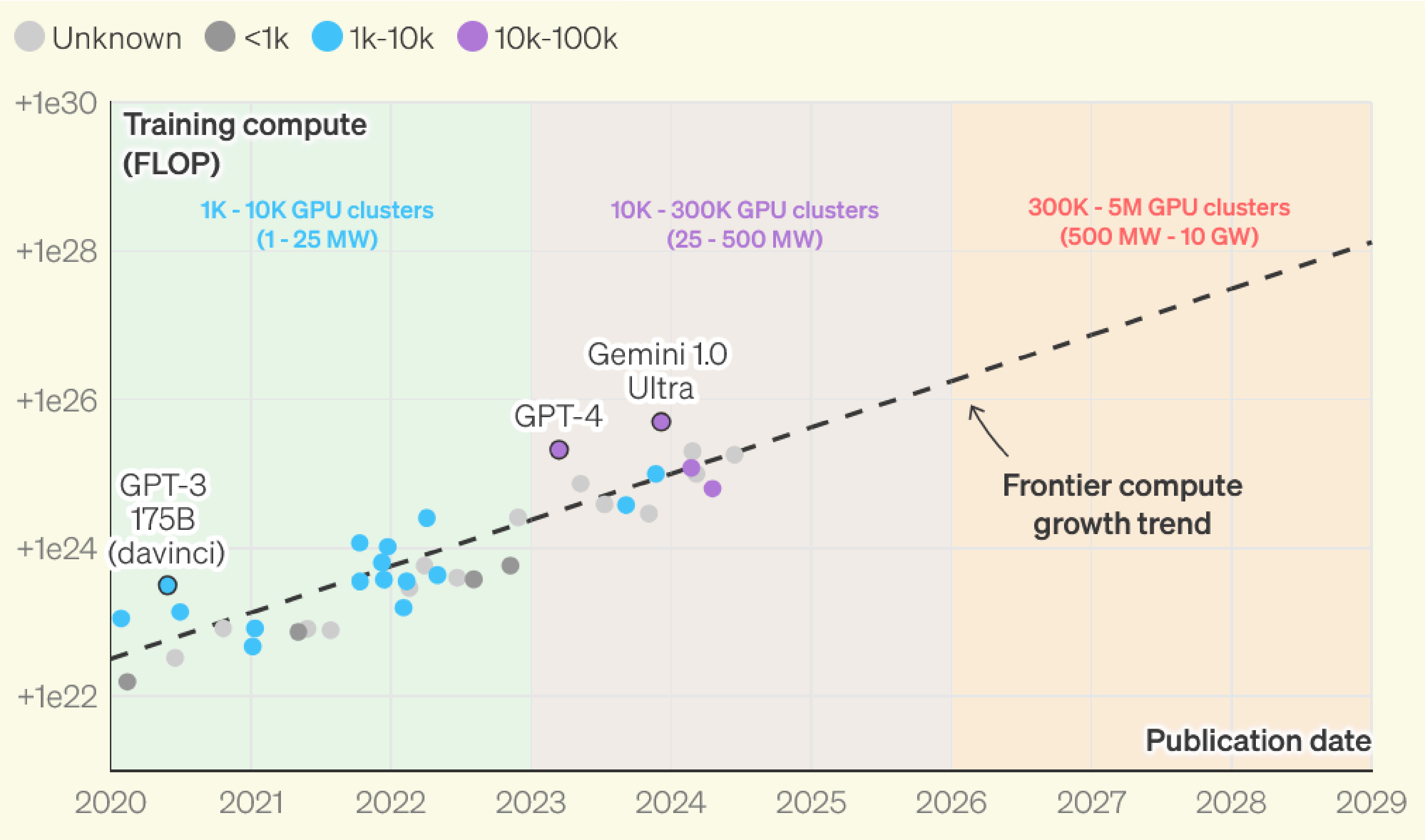


Various estimates for AI-related demand growth

Industry trends: compute demand

- Divergent power demand estimates: different assumptions on AI growth

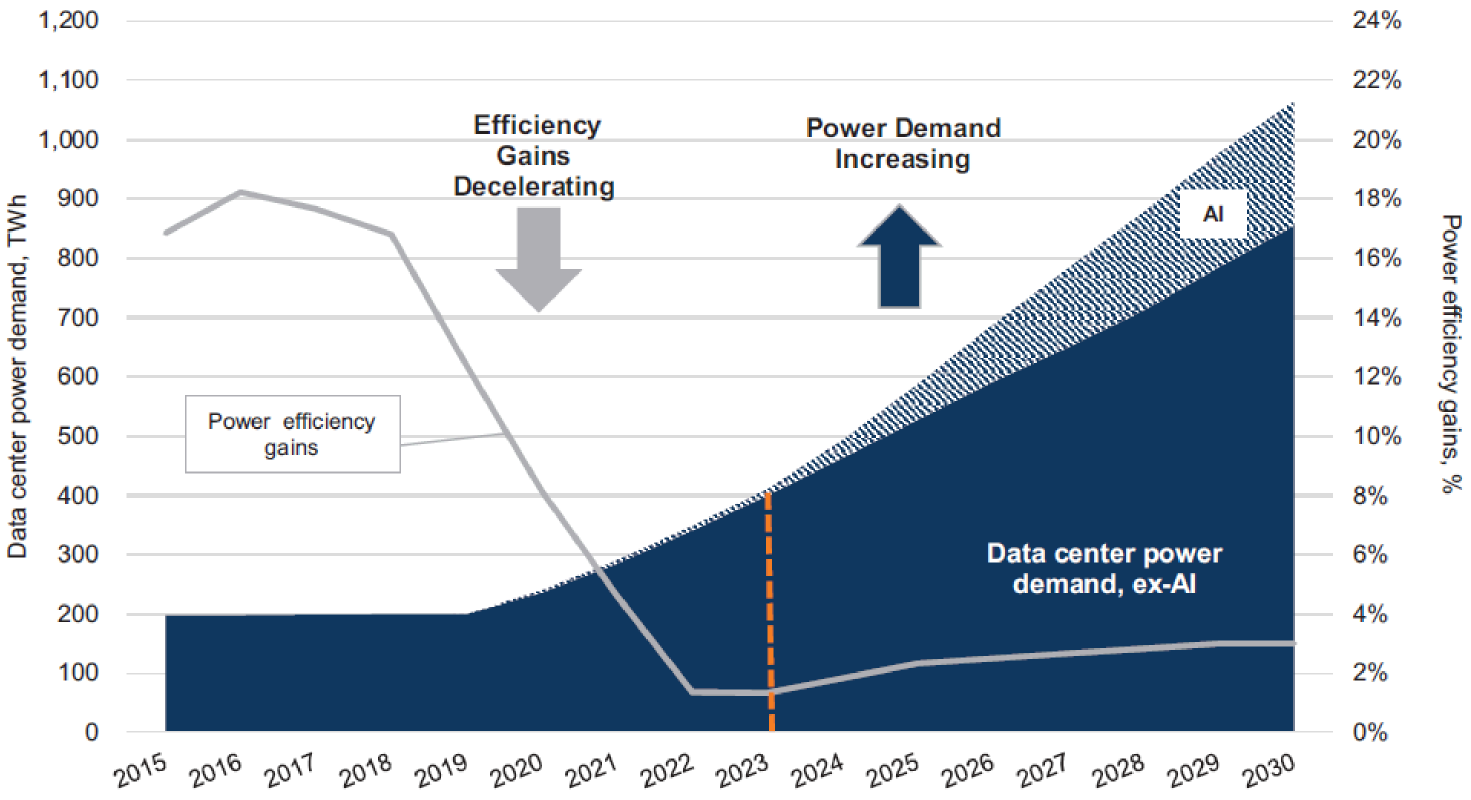
- Recent compute growth for frontier AI models: quintupling scaling law (5x growth per year)
- No consensus on future: hardware improvement, data availability?



Historic trend for frontier model training compute

Industry trends: efficiency gains

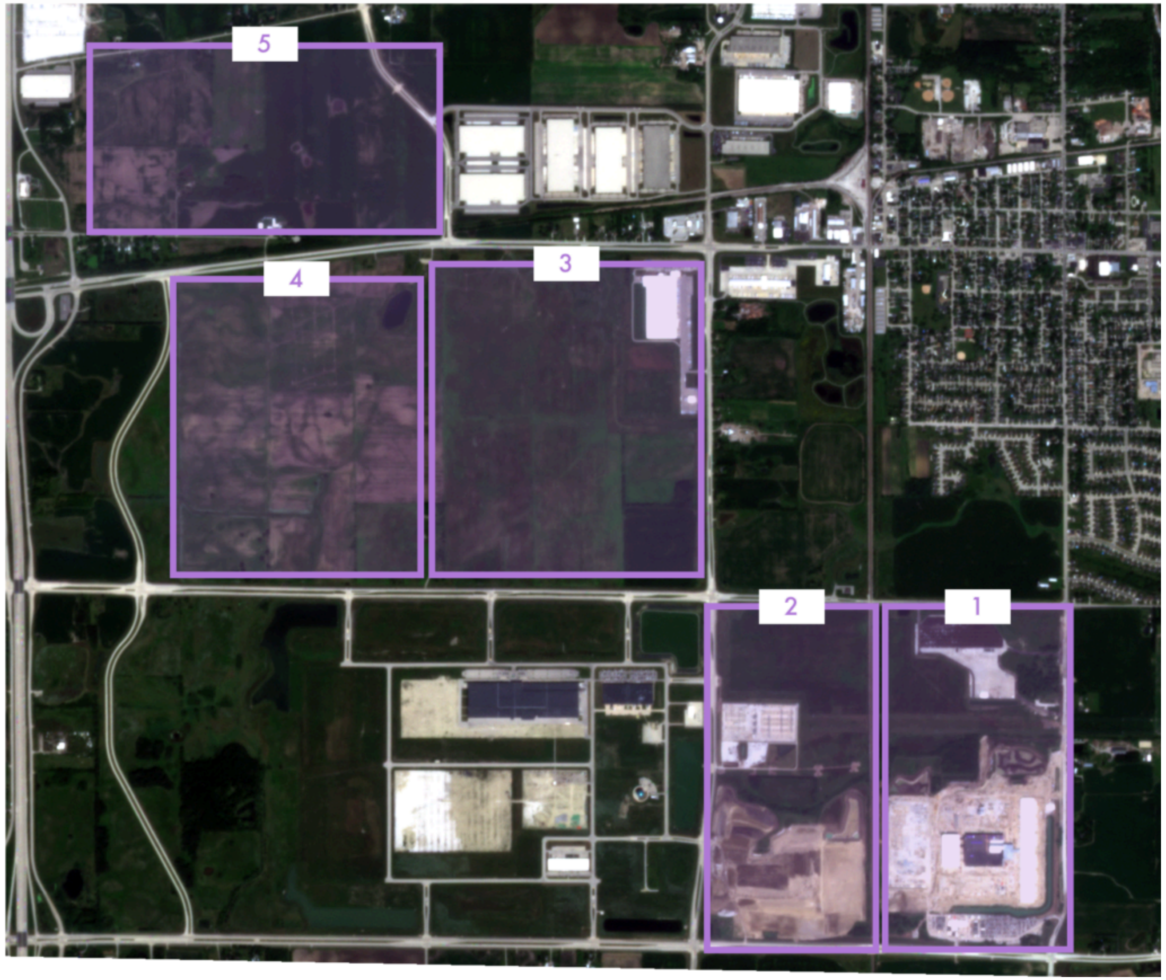
- 2015-2019: constant data center power demand, efficiency gains keep up (~15% yearly)
- Currently: efficiency gains down to 2-4% per year
- Even modern systems (e.g. NVIDIA DGX) cannot keep up with AI training/ inference demand growth



Projected data center power demand and efficiency gains

Industry trends: data center size

- Modern data centers: >10,000 GPUs, >10MW
- Under construction: >100MW
- [Plans:](#) >1GW
- Planned data centers cannot fit in single building/be powered by single grid → campus with wide-area network (WAN)
  - Distributed data centers unlock more AI training → feedback loop!
  - Global fiber cable manufacturing capacity partially reserved by builders!



Planned Microsoft/OpenAI data center campus

Industry trends: energy

- Renewable energy buildout cannot keep up with data center demand growth
- Coal plants meant to be dismantled kept online to power new data centers
- Newly planned buildout of gas plants, pipelines
- Data center builders not waiting for grid interconnection, deploying trailer-mounted gas turbines 'behind-the-meter'
- This is not good

xAI data center with 14 mobile gas turbines

What solutions do we have at our disposal?

Prospective solutions: economies of scale

- Hyperscale data centers more efficient than colocation/internal
- Economies of scale: larger, more energy efficient cooling system vs regular air conditioning; fewer power electronics per computer
- However: hyperscale data center needs to run continuously!
- Also important: where is data center found?
- Strategic regions: desirable geographic/ climatic aspects → reduce need for cooling, water

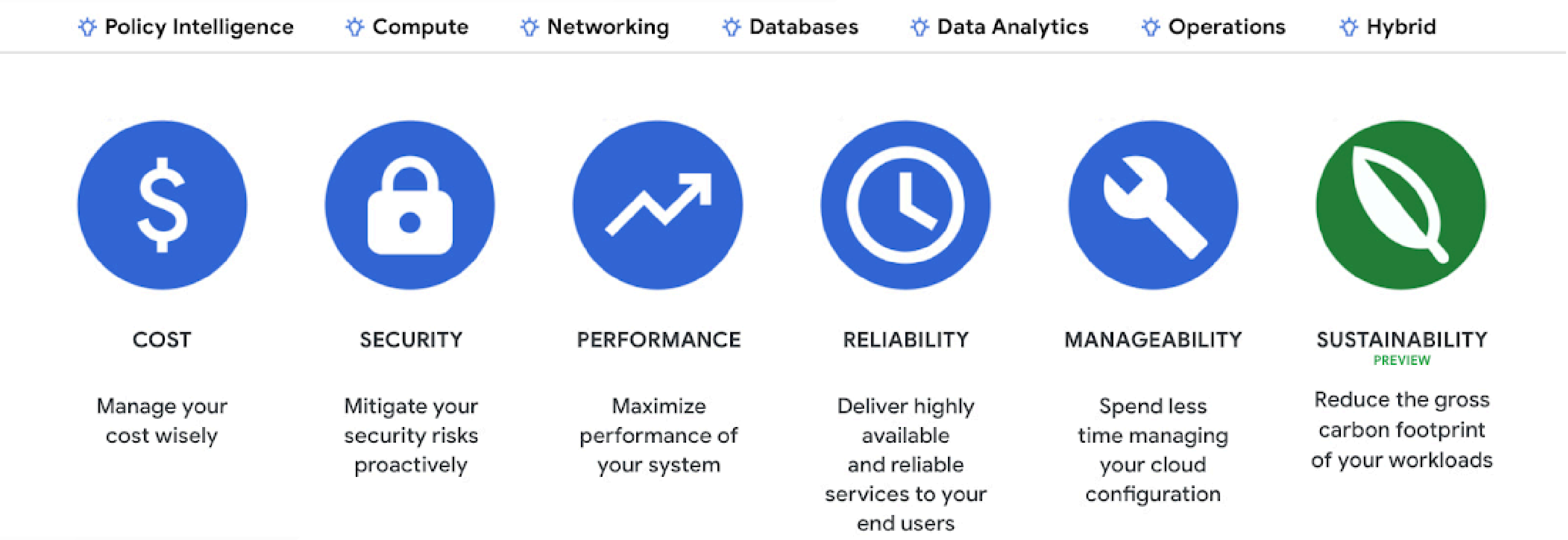
Hyperscale data centres are significantly more efficient than internal data centres

Category	Energy use (Million MWh)	Computing workloads (million)	Water intensity (M_3 MWh_{-1})	Carbon intensity (ton CO_2-eq MWh_{-1})	Water intensity (m_3/workload)	Carbon intensity (Ton CO_2-eq /workload)
Internal	26.90	16	7.20	0.45	12.15	0.75
Colocation	22.4	41	7.00	0.42	3.85	0.25
Hyperscale	22.85	76	7.00	0.44	2.10	0.15

Comparison between hyperscale, colocated, and on-prem data centers

Prospective solutions: cloud strategy

- Cloud strategy: deactivate unused virtual machines, work on-demand
- Maximize server utilization: power electronics designed for efficiency at max load
- Smart scheduling: batch jobs, run code during high utilization periods
- Edge computing: geographically close to user, less energy needed to move data

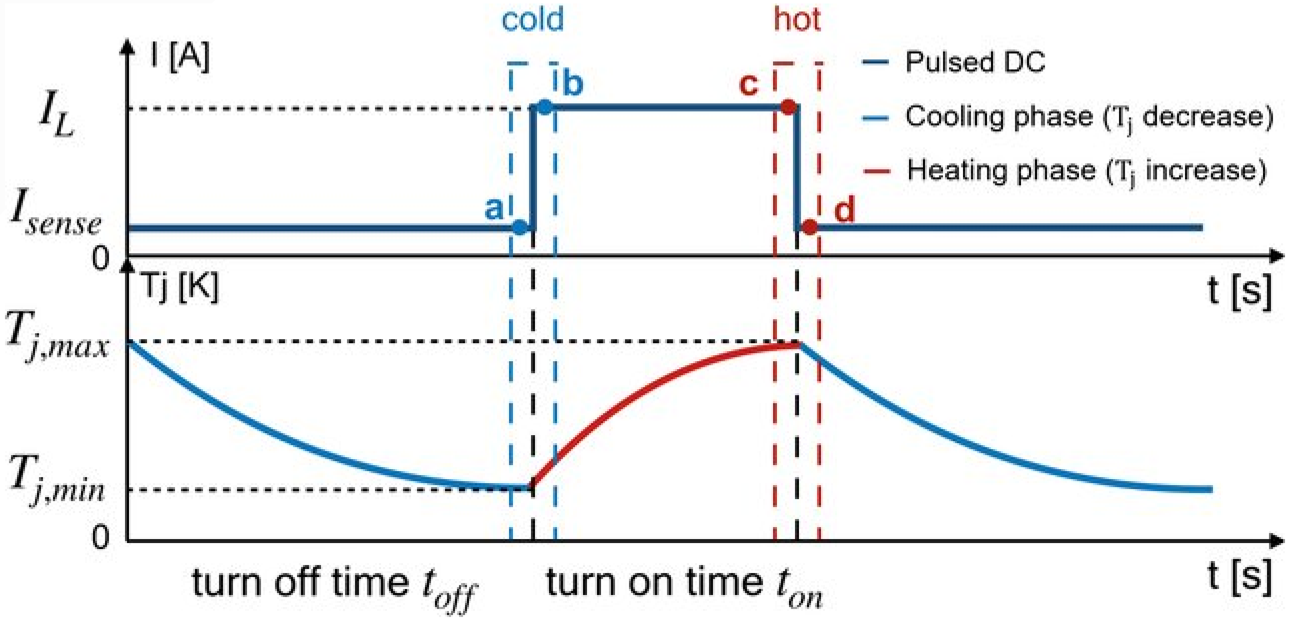


Google Cloud Active Assist smart scheduling

Power cycling

- Smart scheduling: power cycling (on/off) of electronics
- Limited component lifetime: thermal stress due to expansion/compression
- Robust research on lifetime estimation through power cycling, lifecycle assessment of electronics, but none combining the two

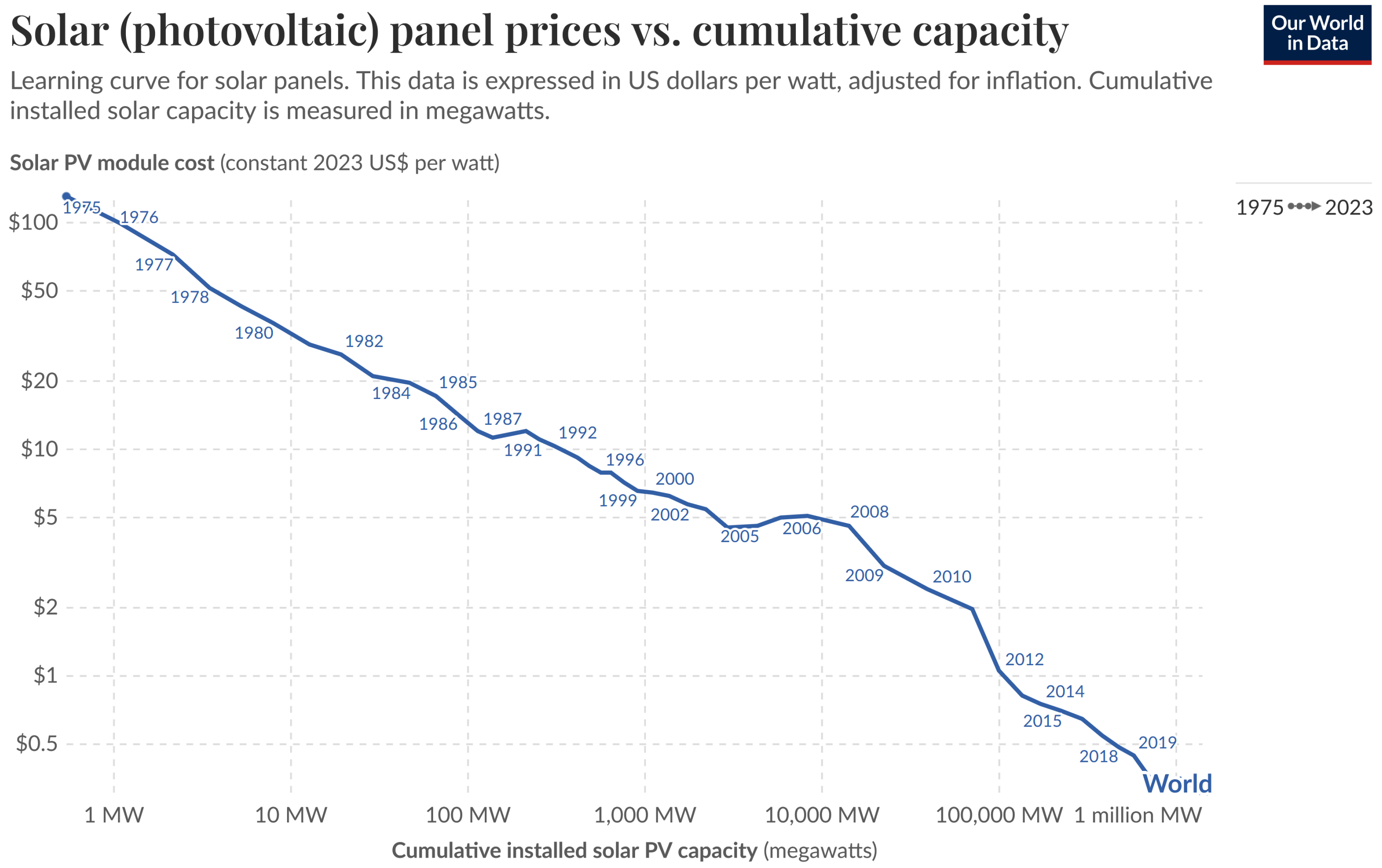
- Complex trade-off: increase energy efficiency, but also frequency of component replacement…
- Master's thesis topic → contact me



Component heating and cooling during power cycling

Prospective solutions: renewable energy

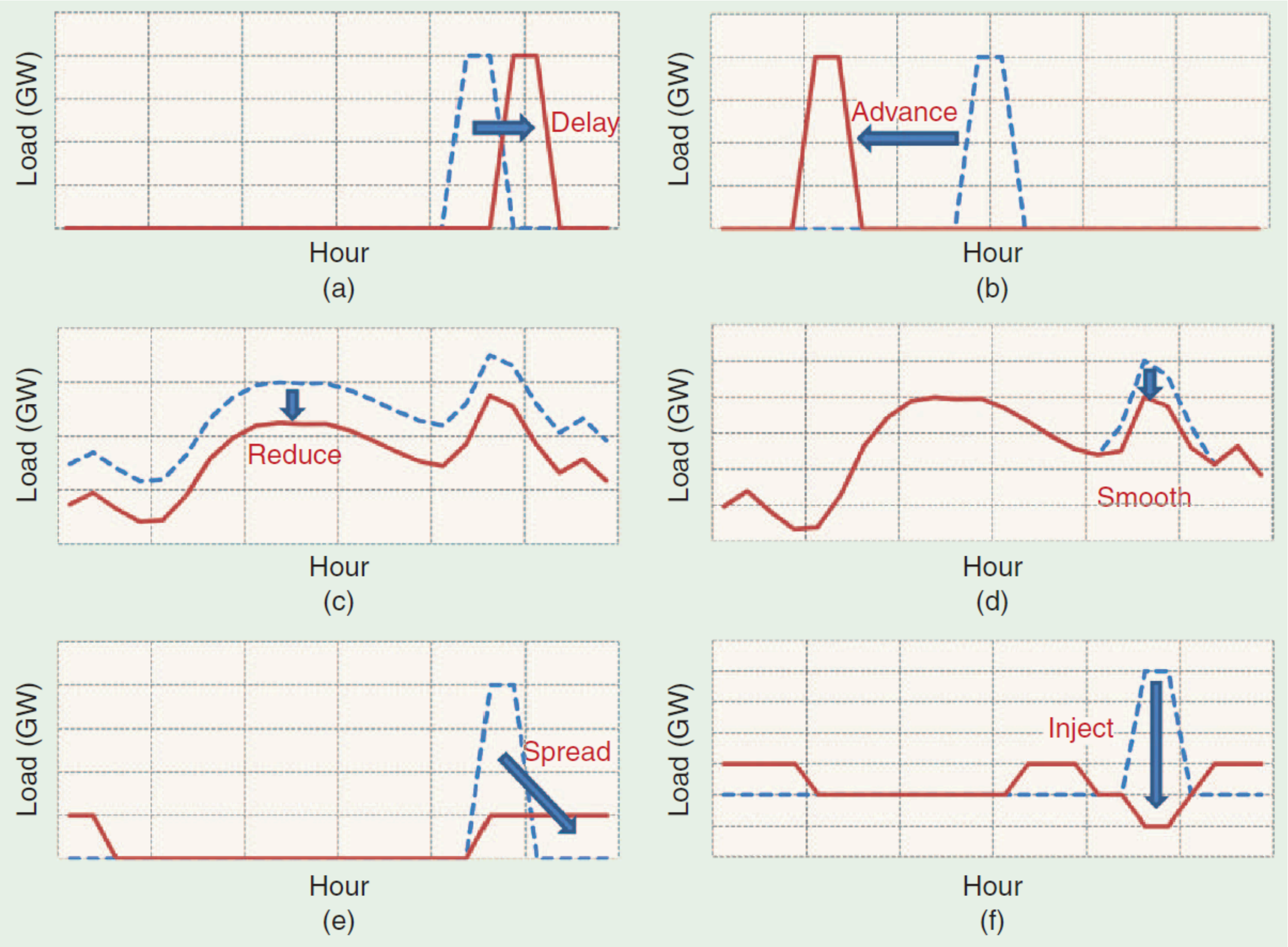
- Renewables (solar, wind etc) do not emit GHG's during operation
- Intermittency: variable output depending on weather conditions
- Favorable learning curves: costs for solar/battery systems falling
- Potential solution: demand response - align high utilization periods with windows of surplus renewable energy



Learning curve for solar PV - note logarithmic abscissa

Prospective solutions: demand response

- Proposal: use parallelizable computing loads as variable demand to balance renewable energy grid
- Example: Monte Carlo compute jobs (low quality of service, allow for pause, no strong privacy/security needs)
- Need to allow jobs to migrate from data center to data center depending on power supply → decentralized platform!



Various demand response strategies

Prospective solutions: server replacement

- Best practices in server maintenance → keep equipment for more than 3-5 years → fewer servers need to be produced
- Problem: latest generation hardware is more energy-efficient
- Complex lifecycle trade-off: operate old, inefficient equipment or throw away and install new, efficient equipment

Prospective solutions: liquid cooling

- Always a trade-off between energy efficiency, water use
- Novel cooling architectures: use liquids (higher heat capacity, 3000x dry air)

- Direct-to-chip (cold plate), immersive cooling: up to 50% more energy-efficient than air cooling
- Problem: direct use of water (closed-circuit?)
- Problem: toxic coolant (PFAS?)
- Higher costs, risks of flooding, difficult maintenance

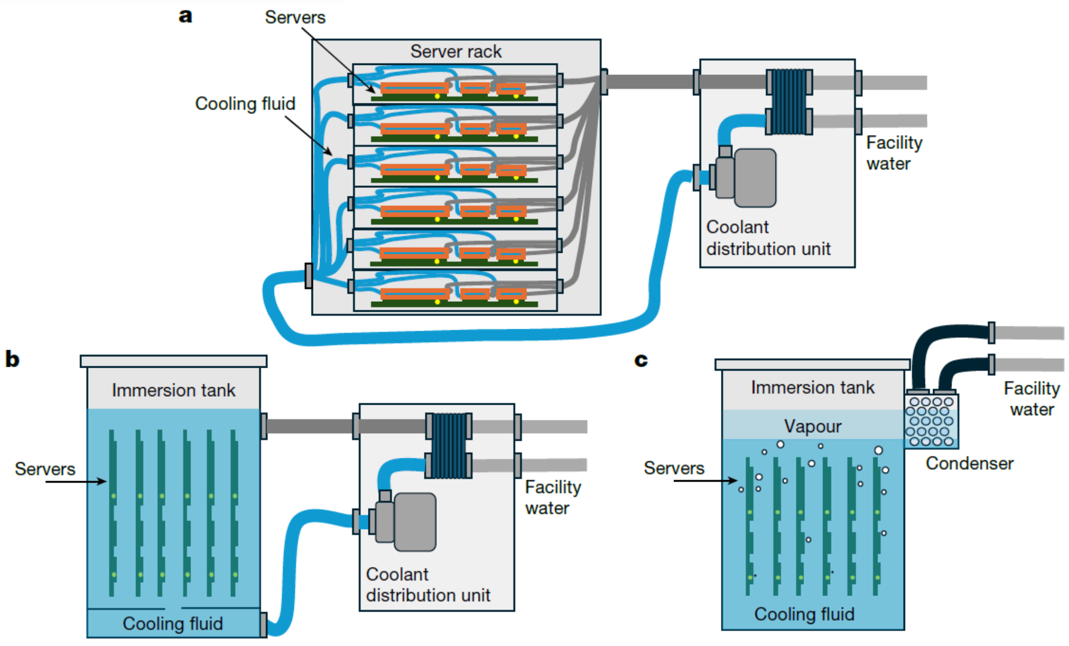


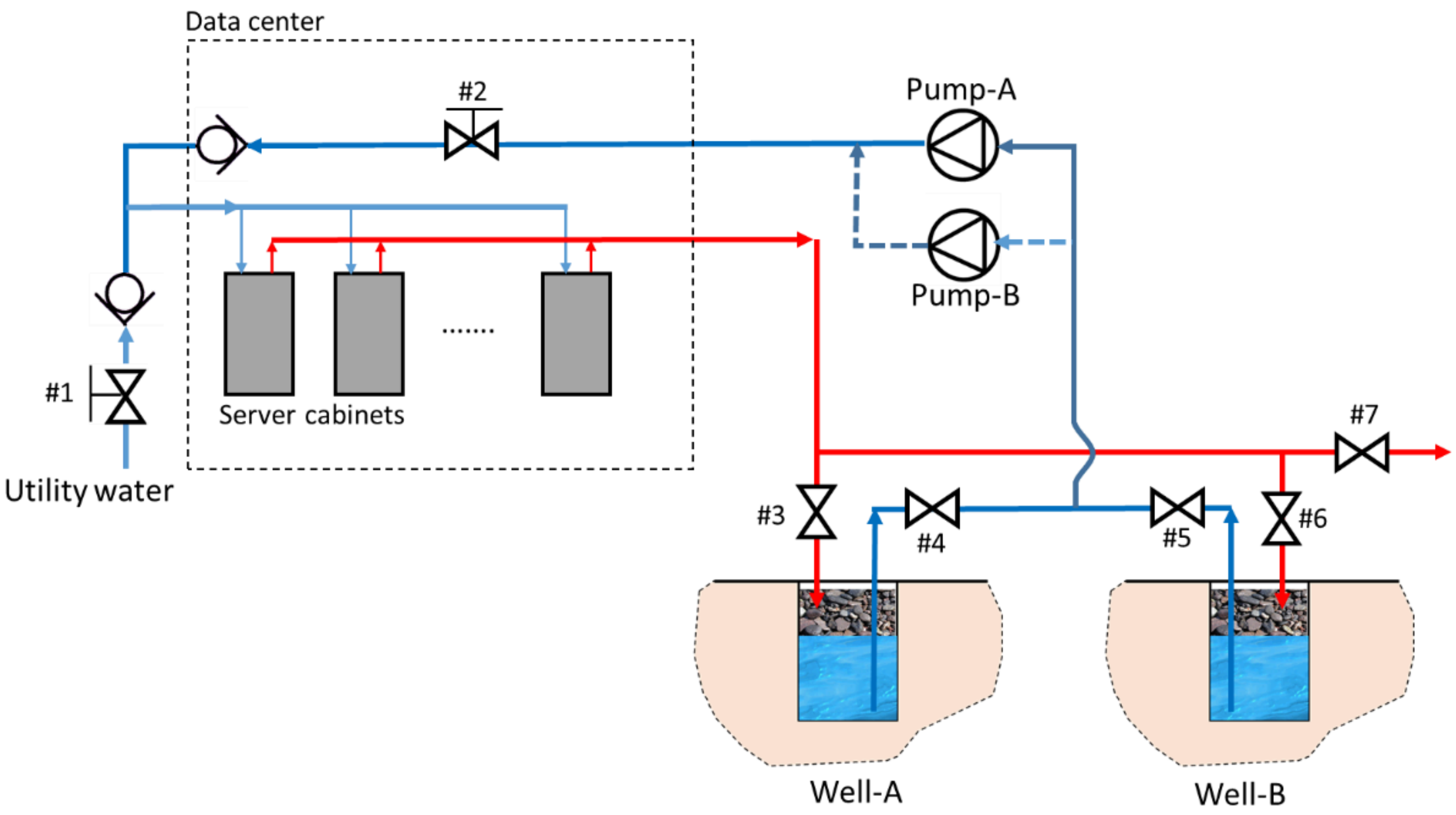
Fig. 1 | Liquid-cooling technologies provide alternatives to air cooling. Schematics showing the functional differences between the cooling technologies. **a**, Cold-plate cooling uses a metal plate with embedded channels for coolant, such as 25% polyethylene glycol and 75% water, to directly absorb and dissipate heat from electronic components with convection[23,24]. **b**, One-phase immersion systems immerse the entire server in a tank of dielectric fluid, such as mineral oil, to remove heat with convection[25]. **c**, Two-phase immersion systems submerge servers in a dielectric fluid that boils at low temperatures, such as a refrigerant, which removes heat with the phase change[23].

(a) cold plate, (b) one-phase and (c) two-phase immersion

Prospective solutions: geothermal cooling

- 15-250 meters deep: ground temperature 10-15 degrees Celsius

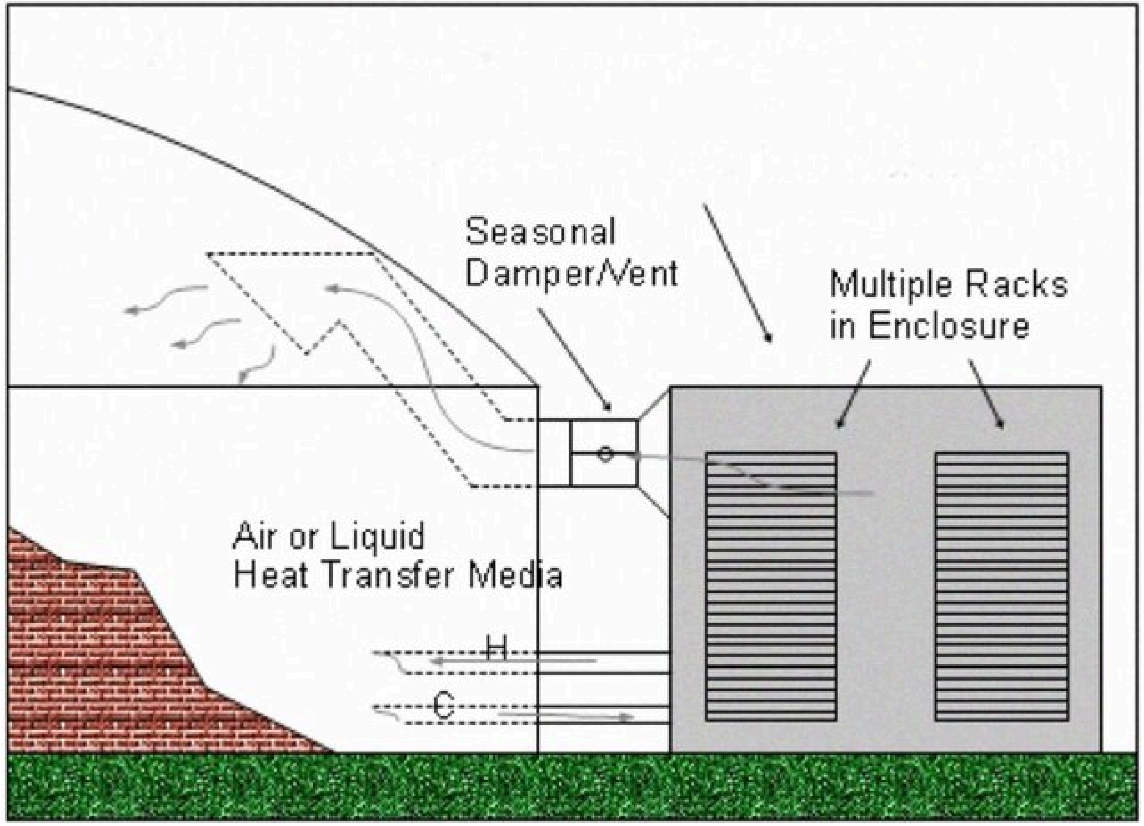
- Dig boreholes and pump warm water from data center to cool down
- Example: Equinix, geothermally-cooled Amsterdam data center
- Lower operational costs, higher installation costs



Geothermal cooling diagram with redundant cooling well

Prospective solutions: waste heat reuse

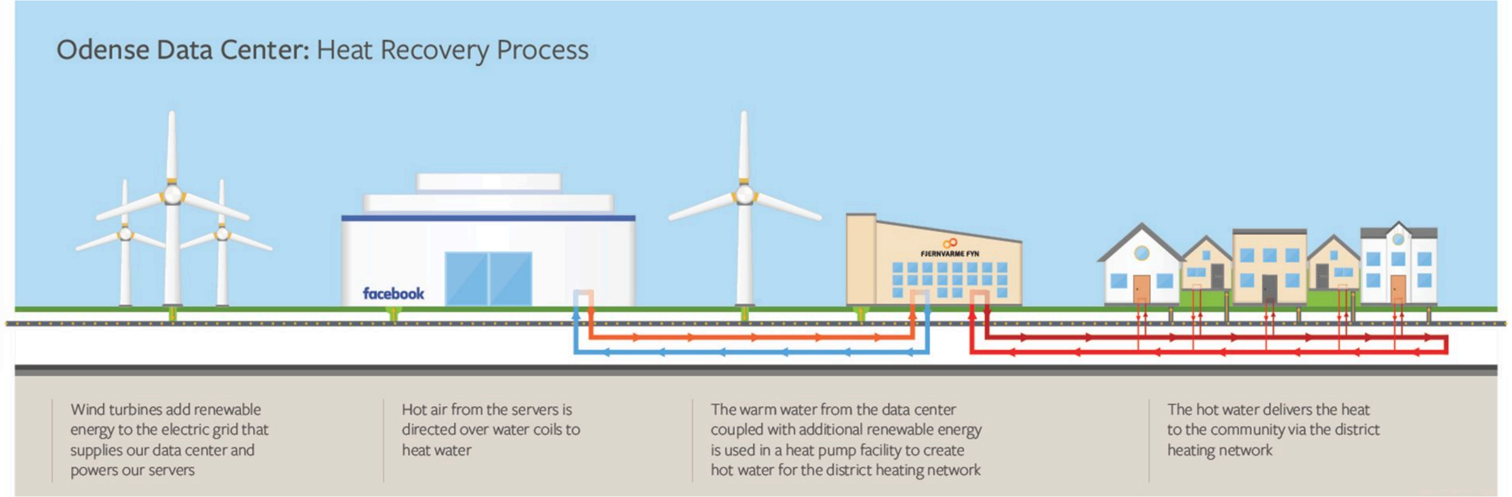
- Free heat → integrate with someone who has demand for heat: industrial symbiosis
- Environmentally opportunistic computing (EOC):
  - create heat where it is already needed
  - exploit cooling where it is already available
  - utilize energy when/where it is cheap
  - use mobility of virtualized services
- Prototype: distributed container-based data center connected to greenhouse
- Computing jobs migrate depending on greenhouse temperature needs
- Dutch startup: BlockHeating



Prototype distributed data center connected to greenhouse

Prospective solutions: waste heat reuse

- Problem: heat demand profile (when do users need heat?)
- Problem: low quality heat produced (30-40 degrees), demand for high-quality heat (>50 degrees)
- Odense, DK: Meta data center recovers waste heat, upgraded to high-quality heat, distributed by heat grid operator to buildings
- Similar developments pursued by other tech companies

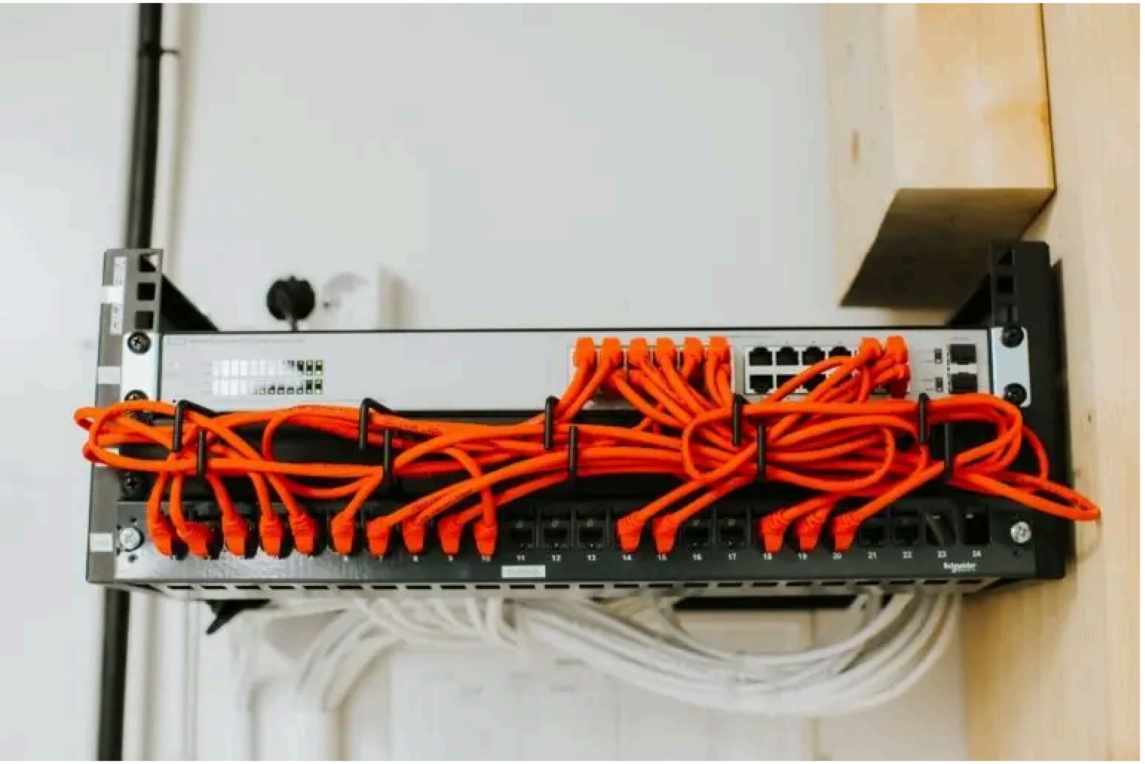


Heat recovery process in Meta's Odense data center

Example: EcoDataCenter

- EcoDataCenter: all-of-the-above approach to sustainability

- Based in Sweden: cold climate, free cooling
- 99.3% renewable energy (75% hydro, 25% wind), some backup use of diesel generators
- Waste heat delivered to district heating, industry needs
- Cooling systems can switch to dry cooling during droughts
- Water tanks used as thermal buffers
- Anything missing?



EcoDataCenter's first building was constructed in wood

Conclusions

- All solutions decrease energy/carbon/material intensity → efficiency gains, vulnerable to rebound effects
- Economic problem: business models of tech companies rely on (exponential) growth in abundance of digital services
- Finite planet → energy constraints to exponential growth?
- Technical solutions insufficient to contain exponential growth → economic/political solutions?

Recap

- Data centers rely on complex computing, cooling, power architectures
- Significant and increasing environmental impacts, estimates need to be made based on external data due to industry opacity
- Growth in compute/power demand, data center size, fossil fuel use projected
- Solutions for reducing energy/carbon/material intensity exist, but are limited and conflict with business models

Exam(ple) questions

1. **A 10MW data center has a yearly energy usage of 60GWh, of which 50GWh are used by actual computing activities, and consumes 250 million liters of water per year. What are its PUE and WUE?**
   a. PUE=1.2, WUE=5L/kWh
   b. PUE=0.83, WUE=5L/kWh
   c. PUE=1.2, WUE=4.17L/kWh
   d. PUE =0.83, WUE=4.17L/kWh
2. **A data center has a PUE of 1.1, a utilization of 70%, and is rated for 20MW at peak load. If it is powered by a coal plant with a carbon intensity of 500gCO2eq/kWh, how many tons of CO2 can be attributed only to the computing activities in a year?**
   a. 6.36 tons of CO2 per year
   b. 7 tons of CO2 per year
   c. 9.1 tons of CO2 per year
   d. 10 tons of CO2 per year

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Feedback

- This is the first edition of the course - we need your feedback!
- Feedback on Lecture 5 slides/notes:

Thank you for your attention!

Next lecture: Sustainable Electronics Design

Thursday, October 16, 15:45-17:30, EEMCS Hall F

- Material basis of ICT systems, recycling
- Sufficient electronics
- **WHAT ELSE?**

Questions?